AMOL WAGHE

AI & Machine Learning Engineer | Generative AI Specialist
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PROFILE SUMMARY

Total 9 years of experience and as AIML and Data Scientist with 4.5+ years of experience specializing in end-to-end AI/ML solution development. Proven expertise in Machine Learning, Natural Language Processing (NLP), Generative AI, and Agentic AI workflows. Strong background in building and deploying scalable AI systems in cybersecurity and enterprise environments.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Regex

ML & Deep Learning: Scikit-learn, TensorFlow, PyTorch, NLP, Computer Vision, YOLOv7

Generative AI: LangChain, LangGraph, RAG, Agentic AI, Vector Databases, OpenAI, Anthropic, Llama

Libraries & Tools: Pandas, NumPy, NLTK, SpaCy, Hugging Face, FastAPI, Flask, Streamlit
Cloud & DevOps: AWS, Azure OpenAI, Docker, Git, GitHub, Linux
Databases: MySQL, MongoDB, MS Excel

WORK EXPERIENCE

Lead Data Scientist

Persistent Systems | Nov 2025- Present Developing RAG based chatbot and Agents for searching clinical oversight project for identifying the study protocols details.

Senior Software Engineer
Candorworks Pvt Ltd | April 2022 – Nov 2025 Developed LLM-powered applications using LangChain, OpenAI, and Llama for email security and threat detection. Built and deployed 5–6 end-to-end email classification models using NLP and deep learning architectures. Designed and implemented RAG-based chatbots for internal and customer-facing use cases. Collaborated with cybersecurity teams to integrate AI models into production pipelines.

ML Engineer(Shift Engineer)
UPSPL Pvt Ltd (Volkswagen) | July 2021 – March 2022

Developed machine learning models for classification and regression using Python and Scikit-learn. Built NLP-based preprocessing and modeling pipelines for text and numerical data. Supported model deployment and monitoring in production environments.

Non-IT Industrial Experience

•**Project Coordinator** Company Name - Jagruti technical services, Pune. (As Project coordinator for HVAC, Utility and Piping). From 1 Sep.2020 to 02 July 2021

• **Sr. Supervisor**- Company Name - FRESENIUS KABI INDIA PVT LTD. Ranjangao Ganpati, Pune. As (HVAC and Utility senior supervisor) From 23 dec. 2019 to 20Aug. 2020.

• **Officer**-Utility and water department (BFS)- Company name: FDC Ltd. Waluj, Aurangabad from 20 July 2019 to 5 dec.2019

• **Sr. Supervisor**- Company Name: FRESENIUS KABI INDIA PVT LTD. Ranjangao Ganpati Pune. (HVAC and Utility senior supervisor) from 28 November 2016 to 16 July 2019.

PROJECTS

●AI-Powered Phishing Text and URL Detection System

Built a real-time phishing detection system using BERT and transformer models. Integrated the model into an email security gateway using Flask API. Technologies: Python, PyTorch, Hugging Face, Flask, NLP.

●RAG-Based Chatbot for Email Threat Manual

Developed a Retrieval-Augmented Generation (RAG) chatbot using LangChain and Pinecone. Enabled querying of threat documentation via FastAPI and WhatsApp integration.

Technologies: LangChain, chromadb, pypdfloader, pdfplumer, FastAPI, OpenAI.

●AI-NLP Based Phishing Email Text Classifier (SkureLabs – NLP Research)

Built a phishing text detection system using advanced AI &NLP techniques.Responsibilities:Developed email body classifier using RoBERTa and BERT for phishing, spam, and benignclassification.Processed mbox and raw email data to prepare NLP-compatible training datasets.Achieved high precision and recall through hyperparameter tuning and dataset balancing.Deployed model in a containerized (Docker-based) REST API, integrated with inbound email filtering.

●Term Deposit Management System: Finance

Empowering Revenue Growth through Customer Segmentation ML Model for TermDeposit Marketing.**Responsibilities**:Developed ML models for customer segmentation to empower revenue growth through Term Deposit Marketing.Evaluated client credit data and financial statements for campaign subscription possibilities. Utilized preprocessing techniques and visualization for insights.Conducted data preparation, analysis, and model training/testing using Python,particularly in the SciKit-Learn ecosystem.